

Homework - On grid: Photovoltaic

Technische Universität Berlin

Chair for Energy Process Engineering and Conversion Technologies for
Renewable Energies

Module: Sustainable energy supply in on- and off-grid systems

The purpose of this homework is the modeling of photovoltaic systems with the Python library *pvlb*. The homework builds on the sample solution of the Photovoltaic exercise. You can do the homework in the provided Jupyter notebook directly.

Inside the homework you need to:

- calculate the photovoltaic system power output using the UI curve model
- calculate the power output and loss of a photovoltaic inverter
- conduct a system sizing and energy yield calculation of a photovoltaic system

Please fill out the missing parts in the Jupyter notebook and answer the questions directly in the notebook. Make sure to print out the results and label clearly. You can answer theory based questions in a Markdown cell within the notebook.

Please ensure that all results are displayed within the notebook with the necessary explanations, we will not run your notebook file on our end. You can also export the notebook as an html file (optional) and upload along with it.

Upload only: Jupyter Notebook (mandatory), HTML export of the notebook (optional)

Dont upload: CSV Files, this PDF file, Or any other extra files